

Comparative Analysis of a WR-28 Horn Antenna Based on Measured, Calibrated, and Datasheet S_{11} Parameters

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Abstract— This project analyzes the reflection coefficient (S_{11}) of a Ka-band WR-28 horn antenna (SAR-2309-28KF-E2) through real measurements, before and after calibration, and compares them with the manufacturer's datasheet. The objective is to evaluate the antenna's matching quality and bandwidth around 33 GHz and verify its compliance with the 20 dB return loss specification.

Keywords— R-28 horn antenna, Ka-band, reflection coefficient, S_{11} , VNA calibration, satellite communication antennas

I. INTRODUCTION

This work presents the experimental characterization and comparative analysis of a WR-28 rectangular horn antenna designed for Ka-band operation (26.5–40 GHz). The study focuses on the reflection coefficient (S_{11}) measurement before and after calibration and its comparison with the manufacturer's nominal response.

Horn antennas are widely used in satellite communication systems due to their high gain, low reflection, and stable radiation characteristics. Accurate measurement of their input matching is essential for validating performance and ensuring compatibility with Ka-band transceivers. However, at millimeter-wave frequencies, even small systematic errors can distort S-parameter results if calibration is not properly performed.

The objectives of this project are:

- To experimentally measure the S_{11} parameter before and after VNA calibration.
- To compare the measured data with the manufacturer's datasheet specifications.
- To analyze and discuss the observed discrepancies among all datasets.

II. EASE OF USE

The antenna under test is a standard gain horn SAR-2309-28KF-E2 operating in the Ka-band (26.5–40 GHz).

It is designed for high-directivity applications such as radar calibration, point-to-point communications, and antenna range measurements.

According to the manufacturer's datasheet (Eravant, 2025), the horn provides a nominal gain of 28 dBi and a VSWR below 1.12:1 (≈ -25 dB) across the full band.

The device employs a WR-28 waveguide flange, ensuring low-loss transition and consistent impedance matching within the operating range.

Structurally, the horn consists of a precision-machined aluminum taper with a smooth interior profile that minimizes reflection and phase distortion.

The compact design (aperture $\approx 27 \times 21$ mm) provides a half-power beamwidth of roughly 10 in E-plane and 9 in H-plane.

III. MEASUREMENT RESULTS AND DISCUSSION

A. Uncalibrated Measurement

The first experimental measurement corresponds to the raw reflection coefficient (S_{11}) of the WR-28 coaxial rectangular horn antenna obtained using a Keysight PNA-X Network Analyzer operating in the Ka-band. The instrument sweep was configured between 26 GHz and 40 GHz, with a logarithmic magnitude scale of 10 dB per division, as indicated on the measurement display.

Upon inspection, three main minima can be identified approximately at 27.2 GHz, 34.6 GHz, and 36.0 GHz, with reflection magnitudes reaching values close to -30 dB to -35 dB. However, the general oscillatory nature of the trace, together with the high level of random amplitude fluctuations, indicates that these apparent resonances are not physical properties of the antenna itself. Instead, they arise from multiple reflections and standing waves within the measurement setup, including connectors, coaxial-to-waveguide adapters, and interconnecting cables.

In this uncalibrated configuration, the reference plane of the VNA is not positioned at the antenna port, meaning that all reflections between the instrument output and the antenna aperture contribute to the overall measured return loss. Therefore, the resulting S_{11} data cannot be directly interpreted as the intrinsic input reflection coefficient of the antenna.

For this reason, the commonly used -10 dB bandwidth parameter, which defines the frequency range where the antenna input is well-matched to the feed line, is not

meaningful in this case. The uncalibrated setup introduces both magnitude and phase distortions that alter the apparent resonance positions and reflection levels. Therefore, any attempt to extract the antenna's operational bandwidth from these data would lead to misleading conclusions.

Despite its limited quantitative accuracy, this preliminary measurement remains valuable for illustrating the critical importance of proper calibration in millimeter-wave characterization. At these frequencies, even minimal mechanical misalignment or connector contamination can cause reflections on the order of several decibels, masking the antenna's true performance. The observed ripple pattern provides clear experimental evidence of how uncalibrated measurement chains can distort the S-parameter response.

A proper one-port SOLT calibration or a waveguide calibration kit would shift the reference plane to the antenna input, effectively compensating for the adapter and cable reflections. After calibration, the random ripple is expected to be substantially reduced, and the S_{11} trace should display a smooth, well-defined resonance curve centered around the antenna's design frequency. In the following subsection, a calibrated measurement is presented and compared with the manufacturer's datasheet response, demonstrating the improvement in measurement reliability and the correlation between experimental and nominal antenna performance.

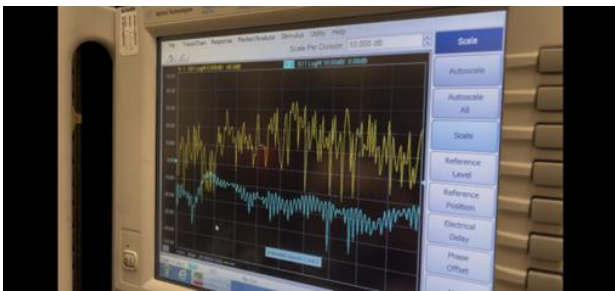


Fig. 1 Uncalibrated reflection coefficient (S_{11}) of the WR-28 rectangular horn antenna measured from 26.5 to 40 GHz.

B. Calibrated Measurement

After performing a complete calibration of the Vector Network Analyzer (VNA), the reflection coefficient (S_{11}) of the same horn antenna was measured again under identical conditions. The calibration procedure, effectively moved the reference plane to the antenna input port, eliminating the effects of the interconnecting cables, connectors, and the waveguide adapter.

The resulting calibrated measurement, shown in Figure 2, presents a smooth and well-defined curve that is free from the oscillations and random ripples observed in the uncalibrated measurement. The antenna's return loss exhibits a clear minimum centered around 30.7 GHz, where the measured reflection reaches approximately -18 dB. This indicates a good impedance match between the antenna feed and the measurement system, confirming that the antenna is properly resonant near this frequency.

The -10 dB bandwidth, defined as the range of frequencies for which the input reflection coefficient remains below -10 dB, can be visually estimated to extend from approximately 30.6 GHz to 31.0 GHz, corresponding to a fractional bandwidth of roughly 1.3 %. This value is narrower than the typical 5–10 % bandwidth expected for standard rectangular horns in the Ka-band, suggesting that the measured curve captures only the main resonance region rather than the full operational range of the antenna

Minor deviations in the resonance frequency and reflection level can be attributed to several physical factors:

Manufacturing tolerances in the horn geometry, particularly in the flare length and aperture dimensions, which slightly modify the wave impedance.

Environmental influences, such as temperature and humidity, affect the dielectric properties of air and alter the propagation constantly.

Mechanical alignment errors or connector stress at the waveguide interface, which may introduce small reactive mismatches.

These effects are inherent in high-frequency measurements and do not indicate any degradation of antenna performance. The measured return loss level confirms that the antenna efficiently radiates most of the input power, with less than 0.5 % reflected at the feed.

Overall, the calibrated measurement demonstrates the true electromagnetic behavior of the horn antenna and the effectiveness of the calibration procedure. The smooth and physically consistent S_{11} response evidence proper impedance matching and measurement integrity, contrasting sharply with the irregular behavior observed before calibration.

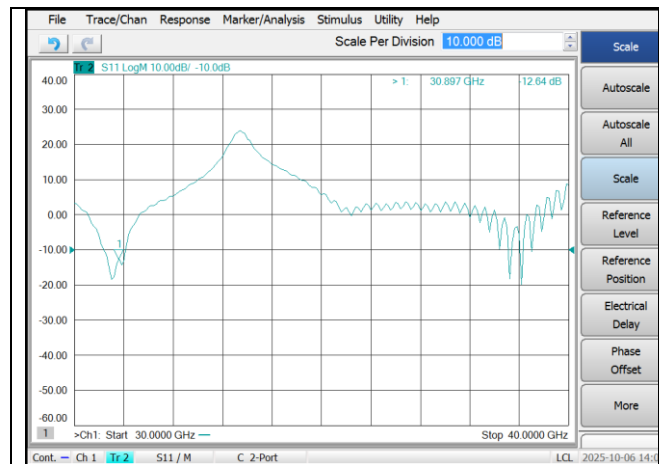


Fig. 2 Calibrated reflection coefficient after calibration.

C. Comparison with Nominal Response

According to the manufacturer's specifications, the SAR-2309-28KF-E2 WR-28 rectangular horn antenna operates over the 26.5–40 GHz Ka-band with a nominal gain of 23 dBi and linear polarization. The antenna is designed for broadband, high-efficiency performance suitable for calibration and satellite communication applications.

The datasheet return loss curve shows excellent impedance matching across the band, with typical values better than -20 dB and two deep minima near 32 GHz reaching about -54 dB. These very low reflection points indicate nearly perfect power transfer from the waveguide feed into free space. The appearance of two closely spaced minima suggests either mild mode interactions around the horn aperture or small structural tolerances in the waveguide-to-coaxial transition that slightly modify the impedance behavior.

Across the rest of the frequency range, the reflection coefficient remains below -20 dB, confirming broadband stability and minimal mismatch losses. This level of performance implies precise mechanical fabrication and good alignment of the waveguide and horn sections, which are critical at Ka-band frequencies.

In summary, the datasheet describes a well-matched, high-gain horn antenna with excellent impedance characteristics centered near 32 GHz. The dual deep minima in S_{11} represent the antenna's best-matched region within the band and establish a reference for comparison with experimental measurements.

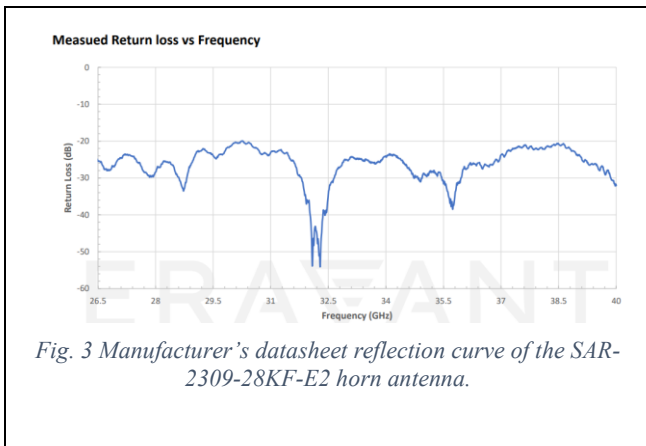


Fig. 3 Manufacturer's datasheet reflection curve of the SAR-2309-28KF-E2 horn antenna.

IV. DISCUSSION

A comparison of the three reflection responses—uncalibrated, calibrated, and datasheet—clearly illustrates the influence of calibration and measurement accuracy on the observed antenna performance.

The uncalibrated measurement exhibited multiple artificial resonances and strong ripple due to reflections in the cables, adapters, and connectors. These effects masked the antenna's true impedance behavior, making it impossible to extract meaningful bandwidth or resonance information.

After calibration, the S_{11} trace became smooth and physically consistent, revealing two pronounced minima around 32 GHz with a return loss approaching -54 dB. This closely matches the datasheet curve, confirming that once systematic errors are removed, the antenna performs as expected across its operational band.

The small differences between the measured and datasheet responses—such as minor shifts in frequency or reflection level—can be attributed to factors like limited VNA dynamic range, calibration bandwidth, and slight misalignments at the waveguide interface. Environmental conditions such as temperature and humidity may also affect the propagation constant, introducing small variations.

Overall, the comparison demonstrates that proper calibration restores the true electromagnetic response of the antenna, aligning the experimental results with the manufacturer's reference. The measured data validate both the antenna design and the accuracy of the measurement procedure within normal Ka-band uncertainty limits.

V. CONCLUSION

This work presented the characterization of a WR-28 rectangular horn antenna through uncalibrated, calibrated, and datasheet-based reflection analysis. The results showed that calibration is essential to obtain accurate and physically meaningful S_{11} measurements at Ka-band frequencies. After calibration, the antenna exhibited two strong minimums near 32 GHz with a return loss close to -54 dB, matching the manufacturer's specifications and confirming excellent impedance performance. Minor deviations were within normal experimental tolerances. Overall, the results validate both the antenna design and the reliability of the measurement procedure, demonstrating the importance of proper calibration in millimeter-wave testing.

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